

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

PREOMICS GMBH AND BIOGNOSYS AG.,
Petitioner,

v.

THE BRIGHAM AND WOMEN'S HOSPITAL, INC.,
Patent Owner

IPR2024-01473
Patent 11,435,360 B2

Before JEFFREY N. FREDMAN, MICHAEL A. VALEK, and
JAMIE T. WISZ, *Administrative Patent Judges*.

VALEK, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Determining Some Challenged Claims Unpatentable
35 U.S.C. § 318(a)

I. INTRODUCTION

PreOmics GmbH and Biognosys AG (collectively “Petitioner”) filed a Petition (Paper 1, “Pet.”), seeking *inter partes* review of claims 1–4, 6, 7, 17, and 22–25 of U.S. Patent No. 11,435,360 B2 (Ex. 1001, “the ’360 patent”). We instituted review. Paper 11, 26 (“Dec.”).

The Brigham and Women’s Hospital, Inc., (“Patent Owner”) filed a Response to the Petition (Paper 18, “Resp.”), Petitioner filed a Reply (Paper 27, “Reply”) and Patent Owner filed a Sur-reply (Paper 33, “Sur-reply”). We heard argument on January 15, 2026, and a transcript is of record. Paper 36 (“Tr.”).

After considering the parties’ arguments and evidence, we find that Petitioner has shown by a preponderance of the evidence that claims 1, 4, 6, 17, 22, and 25 are unpatentable as anticipated by and obvious over Cai.¹ However, Petitioner has not met its burden for its grounds based on Hu² and Zaccaria,³ and the additional claims challenged in those grounds. Our reasoning is explained below.

¹ Cai *et al.*, “Characterization of Carbon Nanotube Protein Corona by Using Quantitative Proteomics,” *Nanomedicine: Nanotechnology, Biology and Medicine* 9 (2013) 583–593, including supplementary data (Ex. 1005) (“Cai”).

² Hu *et al.*, “Nanoparticle Size Matters in the Formation of Plasma Protein Coronas on Fe₃O₄ Nanoparticles,” *Colloids and Surfaces B: Biointerfaces* 121 (2014) 354–361, including supporting information (Ex. 1008) (“Hu”).

³ Zaccaria *et al.*, “Accessing to the Minor Proteome of Red Blood Cells Through the Influence of the Nanoparticle Surface Properties on the Corona Composition,” *International Journal of Nanomedicine* 10 (2015) 1869–1883, including supplementary data (Ex. 1007) (“Zaccaria”).

II. BACKGROUND

A. Real Parties-in-Interest

Petitioner identifies itself as real parties-in-interest. Pet. 1. Patent Owner identifies The Brigham and Women's Hospital, Inc. and Seer, Inc., respectively the owner and exclusive licensee of the '360 patent, as real parties-in-interest. Paper 29, 1.

B. Related Matters

The parties do not identify any other matters involving the '360 patent. Pet. 1; Paper 29, 1.

C. The '360 Patent

The '360 patent issued on September 6, 2022, and claims priority to a series of applications, the earliest of which was filed on December 16, 2016. Ex. 1001, codes (45), (60).

The '360 patent discloses “sensor arrays for detecting biomolecules and methods of use.” Ex. 1001, code (57). This method involves contacting a biological sample with a plurality of particles with different particle types, allowing biomolecules from the biological sample to bind and form “coronas” around them, where the coronas differ based on particle type and may comprise overlapping and distinct proteins. *See id.* at 86:52–87:3. A “biomolecule corona” forms when a particle is covered with a layer of different biomolecules, including proteins. *Id.* at 15:7–22. According to the '360 patent, its “sensor array is able to detect changes in the compositions of the biomolecule corona,” which “allows for determining a unique biomolecule fingerprint” that can “stratify different health and disease states of subjects.” *Id.* at 14:49–58.

Figure 65 of the '360 patent is reproduced below.

FIG. 65

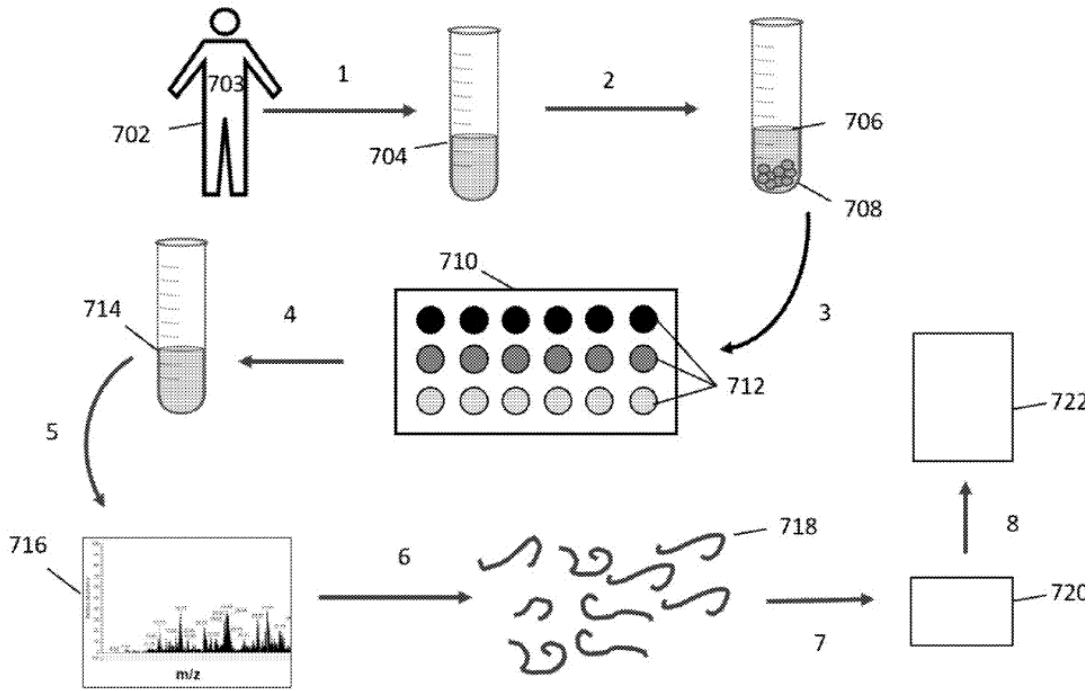


Figure 65 above is a “[s]chematic diagram of a method of distinguishing states of a complex biological sample.” Ex. 1001, 14:25–26. In the first step of the diagramed method, biological sample 704 is collected from a subject 702 expressing a biological state 703 (e.g., a disease state). The plasma 706 is then separated from the blood cells 708 of subjects expressing a biological state 703 in step 2. *Id.* at 21:14–29. In step 3, plasma 706 is incubated with sensor array 710 that includes a plurality of particles 712 with different properties to allow biomolecules (e.g., proteins) in the plasma to bind to the particles. *Id.* at 21:30–36. Next in step 4, biomolecules bound to the particles are isolated in a protein solution 714 to be analyzed “to determine the compositions of the proteins bound to each type of particle 712.” *Id.* at 21:36–42. In Figure 65, for example, the protein solution is characterized by

“liquid chromatography-tandem mass spectrometry (LC-MS/MS) 716” (step 5) and the proteins identified by that analysis are used to determine a biomolecule fingerprint 718 (step 6). *Id.* at 21:43–50.

According to the ’360 patent, “[t]he present sensor array has a sensitivity and dynamic range of ten (10) orders of magnitude in terms of protein detection using mass-spectroscopy approaches,” which is a “greater dynamic range than current assays” using mass spectrometry that “only ha[ve] a dynamic range 4-6 order[s] of magnitude.” Ex. 1001, 20:20–29.

D. Challenged Claims

Independent claim 1 is reproduced below.

1. A method for assaying a biological sample, comprising:
 - (a) contacting the biological sample with a plurality of particles comprising different particle types to permit biomolecules of the biological sample to bind to the plurality of particles and form coronas around the plurality of particles, wherein the coronas corresponding to the different particle types (i) differ based on p article type, and (ii) comprise overlapping and distinct proteins;
 - (b) separating at least a subset of the plurality of particles comprising the coronas from the biological sample by removing the subset of the plurality of particles thereby producing a subset of proteins from the biological sample
 - (c) assaying the subset of proteins of (b) with an instrument to detect, in the subset, proteins in the biological sample at concentrations across a dynamic range comprising at least 6 orders of magnitude, thereby assaying the biological sample.

Ex. 1001, 86:51–87:3. As shown above, claim 1 has three steps:

“contacting” (step 1(a) or limitation 1(a)), “separating” (step 1(b) or limitation 1(b)), and “assaying” (step 1(c) or limitation 1(c)). *Id.* Claims 2–4,

6, 7, 17, and 22–25 depend from and thus incorporate the limitations of claim 1.

E. Asserted Grounds of Unpatentability

The Petition raises four grounds of unpatentability. Pet. 3.

Claim(s) Challenged	35 U.S.C. §⁴	Reference(s)/Basis
1–4, 6, 7, 17, 22–25	103	Hu
1, 4, 6, 17, 22, 25	102	Cai
1, 4, 6, 17, 22, 25	103	Cai
1, 2, 4, 6, 7, 17, 22, 23, 25	103	Zaccaria

In addition to other evidence, Petitioner offers the declaration of Dr. Jessica Vargas (Ex. 1003) in support of these grounds. Patent Owner relies on the declaration of Dr. Michael J. Cima (Ex. 2023).

Each of Petitioner’s grounds rely on one of three references: Hu, Cai, or Zaccaria. We begin our analysis with the two grounds based on Cai before turning to the ground based on Hu, and finally the ground based on Zaccaria.

III. ANALYSIS OF PETITIONER’S GROUNDS

A. Legal Standard

To establish anticipation, each and every element in a claim, arranged as recited in the claim, must be found in a single prior art reference. *Net MoneyIN, Inc. v. VeriSign, Inc.*, 545 F.3d 1359, 1369 (Fed. Cir. 2008);

⁴ The Leahy-Smith America Invents Act, Pub. L. No. 112-29, 125 Stat. 284 (2011) (“AIA”), included revisions to 35 U.S.C. §§ 102 and 103 that became effective prior to the filing of the application that led to the ’360 patent. Therefore, we apply the AIA versions of 35 U.S.C. §§ 102 and 103.

Karsten Mfg. Corp. v. Cleveland Golf Co., 242 F.3d 1376, 1383 (Fed. Cir. 2001). While the elements must be arranged or combined in the same way as in the claim, “the reference need not satisfy an *ipsissimis verbis* test,” i.e., identity of terminology is not required. *In re Gleave*, 560 F.3d 1331, 1334 (Fed. Cir. 2009); *In re Bond*, 910 F.2d 831, 832 (Fed. Cir. 1990). Thus, the dispositive question “is whether one skilled in the art would reasonably understand or infer from a prior art reference that every claim element is disclosed in that reference.” *Eli Lilly and Company v. Los Angeles Biomedical Research Inst. at Harbor–UCLA Med. Ctr.*, 849 F.3d 1073, 1074–75 (Fed. Cir. 2017). Still further, “it is proper to take into account not only specific teachings of the reference but also the inferences which one skilled in the art would reasonably be expected to draw therefrom.” *In re Preda*, 401 F.2d 825, 826 (CCPA 1968).

A claim is unpatentable for obviousness if, to one of ordinary skill in the pertinent art, “the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains.” 35 U.S.C. § 103(a); *see also KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations including (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) when in evidence, objective evidence of nonobviousness. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

B. Level of Ordinary Skill in the Art

Factors pertinent to determining the level of ordinary skill in the art include the type of problems encountered in the art, prior art solutions to those problems, as well as the education level of the inventors and active workers in the field. *Daiichi Sankyo Co. v. Apotex, Inc.*, 501 F.3d 1254, 1256 (Fed. Cir. 2007). The disclosure in the prior art may also reflect an appropriate skill level. *See Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001).

Petitioner contends that a person of ordinary skill in the art (“POSA”) “would have had at least a bachelor’s degree in chemistry, biochemistry, bioengineering, or a related field, and three years’ experience in the field of proteomics, protein analysis, or mass spectroscopy.” Pet. 11 (citing Ex. 1003 ¶¶ 24–27). Petitioner further contends, “[t]his description is approximate – a higher education or skill level might make up for less experience, and vice versa. *Id.* (citing Ex. 1003 ¶ 26). According to Petitioner, its declarant Dr. Vargas, “was at least a POSA as of the ’360 Patent’s earliest possible priority date, December 16, 2016.” *Id.* (citing Ex. 1003 ¶¶ 5–10, 27; Ex. 1004, 1–3).

In the Decision on institution, we found “Petitioner’s description of the level of ordinary skill in the art to be sufficient.” Dec. 8. We noted that the record “supports that a POSA would have an understanding of both proteins and the analytical techniques, e.g., mass spectrometry, that can be used to characterize them.” *Id.* at 9. We also explained that we understood Petitioner’s definition “to encompass these elements” because it requires “a formal degree in chemistry[], which itself entails some knowledge of proteins and related analytical techniques, along with several years’

additional experience in proteomics, protein analysis, or mass spectrometry.”

Id. We also rejected the arguments offered in Patent Owner’s Preliminary Response, suggesting that Dr. Vargas was not qualified because she allegedly lacked experience applying mass spectrometry specifically for protein analysis. *See id.* at 9–10.

Patent Owner does not directly continue those arguments in its Response, but asserts the “petition challenge is improperly constructed from the perspective of a hypothetical person” who could lack sufficient formal training (i.e., only a bachelor’s as opposed to a graduate degree) and experience only in mass spectrometry with no understanding of proteomics. *See Resp.* 65–66.

We disagree with Patent Owner and again find Petitioner’s description of the level of ordinary skill in the art sufficient for the reasons explained in the Decision on institution. *See Dec.* 8–10. As explained before, the formal training required by Petitioner’s description of a POSA includes understanding of both proteins and analytical techniques for studying them. *See Dec.* 9. Neither Patent Owner, nor its declarant, Dr. Cimas, clearly articulates a competing POSA description at trial, but both suggest more formal training, e.g., a graduate degree, is required. *See Resp.* 65–66; Ex. 2023 ¶ 8 (agreeing with our observation that the educational backgrounds of the inventors and authors of Cai, Zaccaria, and Hu “suggests that a graduate degree is more typical for a POSA”) (quoting Dec. 9, n. 5). Even if we were to accept that modification, it would not change our findings herein. Nor does it suggest that Dr. Vargas, who holds a Ph.D. in chemistry from Stanford University and has substantial experience with mass spectrometry and other analytical techniques, is unqualified to testify from the perspective

of one of ordinary skill in the art. *See* Ex. 1004 (Vargas CV). Accordingly, our findings and conclusions would be the same, even assuming the perspective of a POSA possessing the higher level of formal training and specific proteomics experience that Patent Owner suggests is appropriate here.⁵

C. Claim Construction

In an *inter partes* review, claim terms “shall be construed using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. [§] 282(b), including construing the claim in accordance with the ordinary and customary meaning of such claim as understood by one of ordinary skill in the art and the prosecution history pertaining to the patent.” 37 C.F.R. § 42.100(b).

The Petition states “that no express construction of terms is necessary.” Pet. 15. At the institution stage, we agreed that no formal claim construction was necessary, but noted our concerns “regarding the parties’ apparent interpretation of the claim language, or in some instances Patent Owner’s attempt to read-in unrecited limitations.” Dec. 11.

At trial, Patent Owner identifies two terms in claim 1 for express construction: (1) “a biological sample,” and (2) “a subset of the plurality of particles comprising the coronas.” Resp. 13–20. We address the parties’ arguments concerning the construction of these terms below. To the extent

⁵ If anything, the higher level of ordinary skill in the art that Patent Owner advocates would strengthen Petitioner’s arguments that the challenged claims are anticipated by or obvious over the asserted prior art. *See Kinetic Concepts, Inc. v. Smith & Nephew, Inc.*, 688 F.3d 1342, 1366 (Fed. Cir. 2012) (“[I]t is generally easier to establish obviousness under a higher level of ordinary skill in the art.”).

the parties' other arguments raise additional questions of claim interpretation, we address those arguments in the context in which they were raised later in our analysis of Petitioner's grounds.⁶

i. "a biological sample"

Patent Owner contends that the term "sample" is expressly defined in the Specification as "a biological sample or a complex biological sample obtained from *a subject*." Resp. 14 (citing Ex. 1001, 20:33–45). "Thus," urges Patent Owner, "the claimed 'biological sample' is limited by definition to what is obtained from an *individual* subject or patient." *Id.* According to Patent Owner, this construction would exclude the HeLa cell⁷ lysate assayed in Cai as well as the pooled plasma and red blood cell (RBC) samples assayed in Hu and Zaccaria. *Id.*

Petitioner replies, urging that the definition Patent Owner relies upon is for a different term ("sample") and that in the next sentence the Specification "explains what the term 'biological sample' . . . means in the context of the patent stating" that it specifically includes both cell lysates and blood/plasma samples. Reply 1–2. Petitioner characterizes Patent Owner's attempt to exclude the particular samples assayed in Cai, Hu, and Zaccaria as "injecting two artificially narrow, negative limitations in the definition," i.e., that a lysate taken from an immortalized cell line grown in a

⁶ For example, some of Patent Owner's arguments disputing Petitioner's showing for step 1(c) present implicit issues of claim interpretation.

⁷ HeLa cells are an "immortal cell line" commonly used in scientific research. Ex. 1037, 71 (Abstr.); Ex. 1038, 2. This cell line is derived from cervical cancer cells originally taken from "Henrietta Lacks, a patient who died of her cancer" in 1951. *Id.*

laboratory (Cai) and samples pooled from different donors (Hu/Zaccaria) are not obtained from “a subject.” *Id.* at 3–4.

We agree with Petitioner. Both parties refer to the same passage in the Specification to support their competing interpretations, it states:

The term ‘sample’ refers to a biological sample or a complex biological sample obtained from a subject. Suitable biological samples include, but are not limited to, biological fluids, including but not limited to, systemic blood, plasma, serum, lung lavage, cell lysates, menstrual blood, urine, processed tissue samples, amniotic fluid, cerebrospinal fluid, tears, saliva, semen and the like.

Ex. 1001, 20:33–39. As Petitioner points out, the first sentence in this passage does not define the term “biological sample,” but rather the term “sample.” *Id.* According to that sentence, the term “sample” refers to either “a biological sample *or* a complex biological sample obtained from a subject.” *Id.* (emphasis added). Patent Owner’s argument assumes, without explanation, that the phrase “obtained from a subject” modifies both instances of “biological sample” in the first sentence. *Id.* But it need not be read that way. Indeed, that sentence is more naturally read to define “sample” as one of two things—either a “biological sample” or a “complex biological sample” with only the latter being “obtained from a subject.” *Id.*

In addition, the second sentence in the passage above supports Petitioner’s view that the term “biological sample” is not limited to a sample obtained directly from a subject. It provides a broad, non-exhaustive list of “[s]uitable biological samples” that specifically includes the types of biological samples, i.e., “cell lysates,” “systemic blood,” “plasma,” assayed in Petitioner’s cited references. Ex. 1001, 20:36–39. It also lists as an example “processed tissue samples,” which would include samples that have

been modified (i.e., “processed”) from the form in which they were originally obtained. *Id.*; *see also* Ex. 1001, 20:52–57 (describing tissue processing). Given this broad description, we agree with Petitioner that there is no basis in the cited intrinsic evidence for reading in the “negative limitations” Patent Owner seeks to try to distinguish the particular biological samples assayed in Cai, Hu, and Zaccaria. *See* Reply 3–4.

For these reasons, we do not adopt Patent Owner’s proposed construction requiring the recited “biological sample” to be one directly “obtained from a subject” such that it would exclude either the HeLa cell lysate in Cai or the pooled samples from multiple subjects in Hu and Zaccaria.

ii. “a subset of the plurality of particles comprising the coronas”

This phrase appears in step 1(b), which recites “separating at least a subset of the plurality of particles comprising the coronas” formed in step 1(a). Ex. 1001, 86:52–65. As recited in step 1(a), those coronas are formed by contacting the biological sample with “a plurality of particles comprising different particle types . . . wherein the coronas corresponding to the different particle types (i) differ based on particle type, and (ii) comprise overlapping and distinct proteins.” *Id.*

At the institution stage, we observed that there was nothing in the language of claim 1 requiring that all of the contacting in step (a) or the separating in step 1(b) to “occur in the same reaction vessel or at precisely the same time.” Dec. 21–22. We also observed that step 1(b) “does not require that all of the protein coronas formed in limitation 1(a) be separated,” because the claim language recites separating just a “subset” of the particles comprising those coronas and “[g]iven its ordinary meaning,

this language suggests limitation 1(b) could be met by separating particles from only one of the different particle types in limitation 1(a).” *Id.* at 22.

At trial, Patent Owner urges that the “subset” of particles separated in step 1(b) must comprise coronas from each of the different particle types in step 1(a). *See* Resp. 15–20. That is, Patent Owner argues:

The plain and ordinary language of the claims . . . requires *a* subset (singular) ***of the plurality of particles*** comprising ***the coronas*** (plural). The definition for the term “the coronas” is found unambiguously in step 1(a), which requires that “the coronas” correspond “to different particle types” and “differ based on particle type” and “comprise overlapping and distinct proteins.” Ex. 1001, 86:52-87:3. Accordingly, step 1(b) is not satisfied where the corona on the identified subset of particles does not differ based on particle type (i.e., where the subset of particles consists of only one particle type). Here, where the claim recites separating at least a subset (singular) of the plurality of particles comprising “the coronas” (plural), the plain language of the claim requires a single subset of particles that comprises “the coronas,” as recited in the claims.

Id. at 15. Thus, while acknowledging that “limitation 1(b) does not require separating literally every single one of the particles” in step 1(a), Patent Owner contends the subset in step 1(b) must “comprise coronas (plural) defined in the claims as differing on particle type and comprising overlapping and distinct proteins.” *Id.* at 17 (internal quotations omitted).

Petitioner replies, urging that our prior observation that the plain and ordinary meaning of step 1(b) would encompass separating a subset of particles of just one of the different particle types in step 1(a) is correct. *See* Reply 9–12 (urging that “the Board already expressly rejected Patent Owner’s attempt to read limitations” into step 1(b) “and the result should be no different” now). Petitioner characterizes Patent Owner’s claim

construction argument as an attempt to import a “negative limitation,” i.e., that “the claimed subset excludes subsets that contain only one particle type,” that “fails as a matter of plain English and lacks specification support.” *Id.* at 9.

We agree with Petitioner. Step 1(a) recites forming coronas around a plurality of particles of “different particle types.” Ex. 1001, 86:53–60. Step 1(b) recites separating a “subset of the plurality of the particles comprising the coronas.” *Id.* at 86:61–65. As Petitioner correctly observes, as a matter of ordinary English parlance, “there is no requirement whatsoever that the ‘subset’ have all of the same characteristics as the set from which it is separated.” Reply 10. By way of analogy, consider a jar containing a plurality of coins of different coin types, e.g., nickels and dimes. One can separate out a “subset” of that plurality of coins (plural) by removing any two coins from the jar—this is true even if the two coins are both nickels or both dimes. In the same way, nothing in the ordinary meaning of the claim language requires the subset in step 1(b) to include particles with coronas representative of each of the different particle types in step 1(a). Nor has Patent Owner identified any other evidence that supports its attempt to read such a requirement into the claims. *See* Resp. 16, 18 (citing expert testimony that is not inconsistent with the broader, ordinary meaning of the claim language and, even if it was, would not be a proper basis for limiting the same).

For these reasons, we reject Patent Owner’s interpretation of “a subset of the plurality of particles comprising the coronas” in step 1(b) as limited to a subset of *each* type of particle comprising the coronas formed in step 1(a).

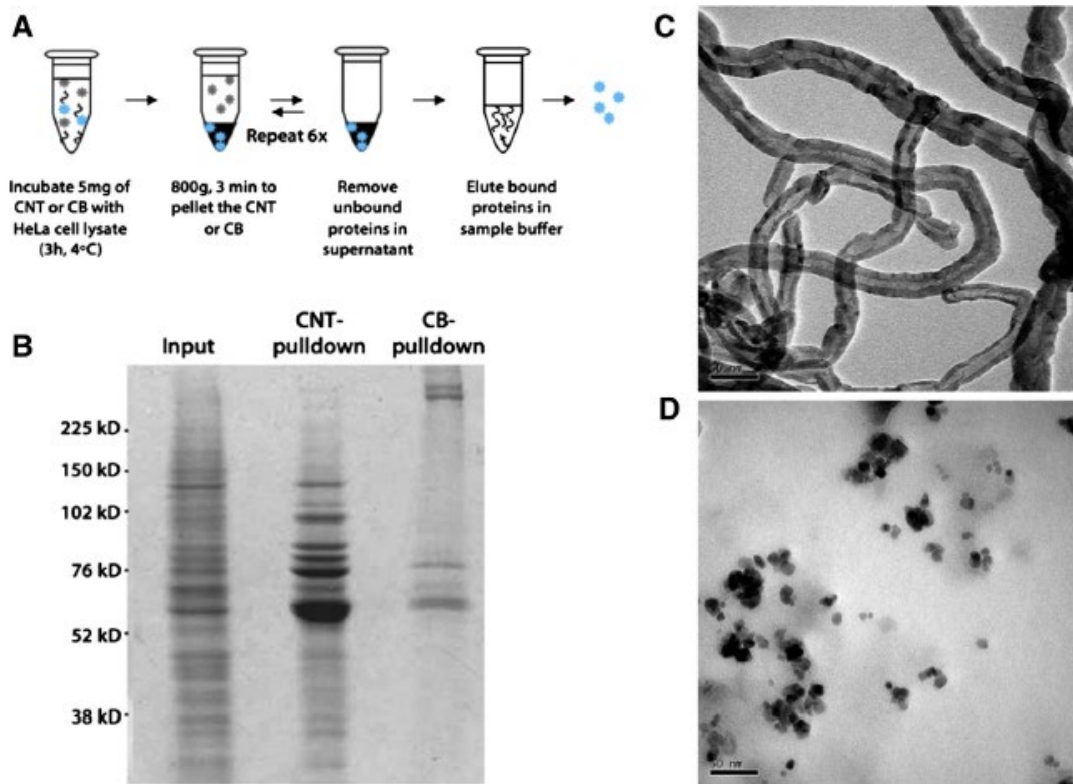
D. Asserted References

i. Cai

Cai is a journal article bearing a September 2012 publication date. Ex. 1005, 1; *see also* Ex. 1012 ¶¶ 26–33 (supporting qualification as prior art). It is undisputed that Cai, including the supplementary data in Exhibit 1005, is prior art to the challenged claims.

According to Cai, “[t]he protein corona of a nanomaterial is a complex layer of proteins spontaneously and stably formed when the material is exposed to body fluids or intracellular environments.” Ex. 1005, Abstr. Cai describes a study “utili[zing] stable isotope labeling by amino acids in cell culture (SILAC)-based quantitative proteomics to characterise the binding of human cellular proteins to two forms of carbon nanoparticles: namely multi-walled carbon nanotubes (MWCNTs) and carbon black (CB).” *Id.*

In Cai’s study, HeLa cell lysates were incubated with CB or carbon nanotubes (CNTs), and in some instances with “CNTs of different sizes.” Ex. 1005, 584. The proteins bound to those particles were then eluted for further analysis. *Id.* Figure 1 of Cai, reproduced below, illustrates this process.

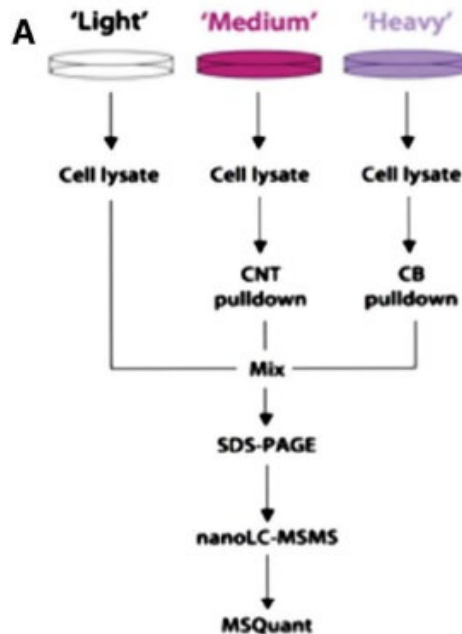


Ex. 1005, 586. Figure 1A (top left above) schematically shows the incubation of a cell lysate with carbon nanotubes or carbon black, followed by elution of the proteins that bind to the particles. *Id.* at 585. Figure 1B (bottom left) shows separation of the bound proteins by SDS-PAGE⁸ in a “CNT-pulldown” and “CB-pulldown.” *Id.* at 586. The figure also shows TEM [transmission electron microscopy] images of the carbon nanotubes (C) and carbon black (D) on the right. *Id.*

Cai describes further testing “to compare the types of proteins that bound to . . . MWCNT with those that bound to CB.” Ex. 1005, 586. “Cell lysates were prepared from three populations of HeLa cells that had been

⁸ “Sodium Dodecyl Sulfate-Polacrylamide Gel-Electrophoresis (SDS-PAGE)” is a technique “widely used for protein separation.” Ex. 1003 ¶¶ 34, 179.

metabolically encoded using three different SILAC media ('light', 'medium' and 'heavy', Figure 3, *A* and Methods).” *Id.* Figure 3A of Cai, reproduced below, illustrates this process.



Id. Figure 3A above schematically shows a “CNT pull-down . . . performed on the ‘medium’ lysate, and a CB pull-down . . . performed on the ‘heavy’ lysate.” *Id.* “Equal volumes of the proteins eluted from these pull-down experiments were mixed” and “spiked with an equal amount of unfractionated ‘light’ lysate.” *Id.*; *see also id.* at 584 (referring to the light-labeled lysate as a “control”). “The mixtures were then separated by SDS-PAGE, and analyzed with the ‘gel-LC-MS/MS’ approach,” i.e., liquid chromatography (LC) and mass spectrometry (MS). *Id.* at 586–587. The obtained spectra were then “quantified by the MSQuant programme” to identify unique peptides and “[t]he abundance distribution of each identified peptide among the ‘light’, ‘medium’ and ‘heavy’ categories.” *Id.* at 587.

Cai discloses the “empirical peptide abundance index (emPAI),” which it describes as “a semi-quantitative indication of the overall

abundance of protein in the mixture,” for each protein detected in this assay. Ex. 1005, 587 (citing Ishihama⁹); *see also* Ex. 1005, 12–17 (Supplementary Table 2 reporting emPAI values). Cai discloses that “[t]he relative abundance of the MWCNT-binding proteins” identified through this analysis “spread over 8 orders of magnitude.” Ex. 1005, 587. Cai further discloses that “while a small number of CB-binding proteins also bind to MWCNTs . . . most of the proteins with high CB-binding efficiencies bind to MWCNTs only marginally or not at all,” which indicates that these nanoparticles have both different protein binding capacities and different protein binding specificities. *Id.* According to Cai, this demonstrates “that CNTs are able to selectively enrich some specific proteins from protein mixtures.” *Id.* at 592.

ii. Hu

Hu is a journal article bearing a June 2014 publication date. Ex. 1008, 354; *see also* Ex. 1012 ¶¶ 16–25 (supporting qualification as prior art). It is undisputed that Hu, including the supporting information in Exhibit 1008, is prior art to the challenged claims.

Hu describes a study investigating “whether nanoparticle size affects the formation of protein coronas around Fe₃O₄ [nanoparticles].” Ex. 1008, Abstr. In Hu’s study, Fe₃O₄ nanoparticles (NPs) of varying sizes were mixed with human plasma to form nanoparticle-protein complexes that “were collected by a magnet and washed at least 3 times to remove the loosely

⁹ Ishihama *et. al.*, “Exponentially Modified Protein Abundance Index (emPAI) for Estimation of Absolute Protein Amount in Proteomics by the Number of Sequenced Peptides per Protein,” *Molecular & Cellular Proteomics* 4.9 (2005) 1265–1272 (Ex. 1018) (“Ishihama”).

bound and unbound proteins.” *Id.* at 355. The complexes were then resuspended and divided into two parts, wherein one part was purified by SDS-PAGE “to illustrate the binding patterns of the plasma proteins on the Fe₃O₄ NPs.” *Id.* “The other portion was analyzed” with a “nanoparticle digestion shotgun proteomics method,” involving digestion with trypsin followed by analysis of “[t]he digested peptides from the human plasma protein corona” using LC-MS/MS. *Id.* A software program was then used to compare the MS/MS spectral data with a database to identify and quantify unique proteins. *Id.* at 356.

According to Hu, the results from these experiments show that “particle size does play an important role in the formation of plasma protein coronas on Fe₃O₄ NPs,” affecting both protein composition and abundance. Ex. 1008, Abstr.; *see also id.* at 359, Fig.4 (Venn diagram identifying overlap and non-overlap of proteins bound by different sizes of Fe₃O₄ NPs).

iii. Zaccaria

Zaccaria is a journal article bearing a March 2015 publication date. Ex. 1007, 1869; *see also* Ex. 1012 ¶¶ 12–15 (supporting qualification as prior art). It is undisputed that Zaccaria, including the supporting information in Exhibit 1007, is prior art to the challenged claims.

Zaccaria describes a study of protein coronas formed on “three [different] kinds of superparamagnetic NPs.” Ex. 1007, Abstr. In Zaccaria’s study, “each kind of NPs were individually incubated” with a red blood cell (RBC) lysate prepared from human blood samples. *Id.* at 1871. “Then, NPs were attracted to one side of the vial by an external magnet, and supernatant was removed.” *Id.* Bound proteins were then purified by SDS Page and analyzed using mass spectrometry (i.e., nano-LC-MS/MS) to identify and

quantify the proteins in the coronas of each of the three different kinds of NPs. *Id.* at 1871, 1873–1874. Zaccaria reports that the different types of NPs bound different proteins. *See, e.g., Id.* at 1873, Fig. 2A–C.

E. Anticipation by Cai

Petitioner contends claims 1, 4, 6, 17, 22, and 25 are anticipated by Cai. *See* Pet. 46–61. Patent Owner disputes Petitioner’s contentions, raising arguments against Petitioner’s showing for claim 1 that also apply to dependent claims 4, 6, 17, 22, and 25. Resp. 21–34.

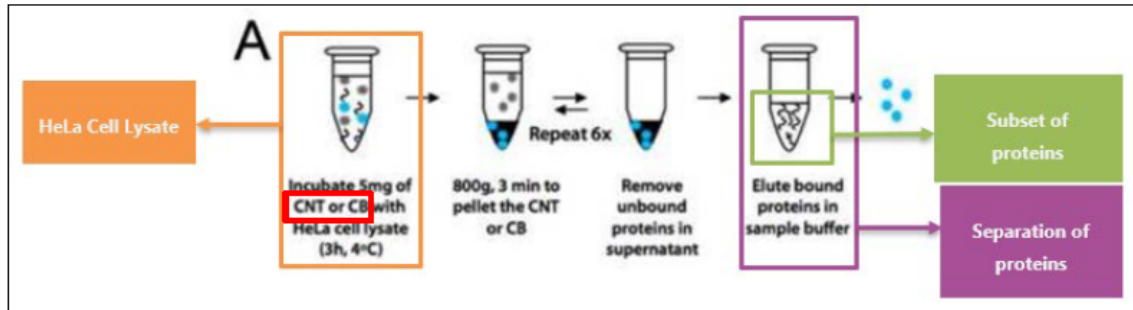
For the reasons explained below, we find that Petitioner has shown by a preponderance of the evidence that Cai anticipates these claims. We begin our analysis with claim 1.

i. Claim 1

Petitioner asserts that “Cai’s method of analyzing the protein lysate from HeLa cells using various CNTs alone or with CB discloses the claimed method.” Pet. 48 (internal quotations omitted). In particular, Petitioner relies on Cai’s experiments involving MWCNT and CB, which it maps to step 1(a)’s recitation of contacting a “plurality of particles comprising different particle types” with the biological sample to form coronas comprising “overlapping and distinct proteins.” *See id.* at 50–57 (citing evidence).¹⁰ According to Petitioner, after the lysate is incubated with the

¹⁰ Petitioner presents this as an “alternative” argument to its separate reliance on Cai’s experiments involving different types of CNTs “alone constituting the claimed ‘*plurality of particles comprising different particle types*’” in limitation 1(a). Pet. 50. However, because Petitioner’s showing for the other claim limitations is largely focused on Cai’s disclosure as it relates to

MWCNT or CB, Cai discloses that the proteins bound in the coronas are eluted to produce a subset of proteins that are then further analyzed by SDS-PAGE. *Id.* at 57–58 (citing Ex. 1005, 585, Fig. 1A; Ex. 1003 ¶¶ 223–224). Petitioner illustrates its position with an annotated version of Figure 1A of Cai, reproduced below.



Petitioner’s annotated version of Figure 1A above includes an orange box around the incubation step on the far left, identifying the HeLa cell lysate (biological sample) that is incubated with the “CNT or CB” (plurality of particles) identified with a red box to form coronas of bound proteins. Petitioner annotates the “elute bound proteins” step on the far right with a purple box and a smaller green box around the proteins themselves, which it contends constitute a “subset of proteins” as recited in step 1(b). *Id.* at 57 (quoting Ex. 1005, 585 (“The proteins bound stably to [the MWCNT and CB particles] were eluted after extensive washing and then analyzed by SDS-PAGE.”)).

For step 1(c), Petitioner argues that Cai “discloses multiple methods of assaying the isolated protein coronas taken from the bound carbon nanoparticles, including gel electrophoresis, LC-MS/MS with SILAC

MWCNT and CB (*see* Pet. 51–60), we view it as the stronger of Petitioner’s arguments and focus on it here.

encoding and Western Blotting.” Pet. 59. In particular, Petitioner relies on the results from the LC-MS/MS with SILAC encoding testing that Cai states show “[t]he relative abundance of the MWCNT-binding proteins spread ***over 8 orders of magnitude***.” *Id.* at 60 (quoting Ex. 1005, 587) (emphasis in original). According to Petitioner, this result demonstrates “that Cai’s MS analysis detected protein concentrations in the subset across a dynamic range comprising the claimed ‘*at least 6 orders of magnitude*,’ which is further confirmed by the concentrations [of proteins] detected in Cai.” *Id.* (citing Ex. 1003 ¶¶ 97, 229).

We credit Petitioner’s showing above and agree that Cai discloses a method comprising all of the steps in claim 1. Beginning with step 1(a), Cai discloses contacting two different particle types, i.e., MWCNT and CB, with HeLa cell lysate (i.e., a biological sample) to form protein coronas around those particles. Ex. 1003 ¶¶ 199–206; Ex. 1005, Abstr. (describing MWCNT and CB as distinct “forms of carbon nanoparticles” that bind “vastly different sets of proteins”). Cai’s results show the coronas formed around the MWCNT and CB particles comprise overlapping and distinct proteins as recited in step 1(a). Ex. 1003 ¶¶ 216–222; Ex. 1005, 587 (“a small number of CB-binding proteins also bind to MWCNTs”). A subset of the proteins from the coronas on these particles is then prepared by eluting the bound proteins and analyzing them with SDS-PAGE in a MWCNT and CB “pull-down,” which reads on step 1(b). Ex. 1005, 586 (Fig. 1A–B depicting scheme for “pull-down experiment” and describing other “pull-down experiments” used with SILAC media in Fig. 3), 588 (Fig. 3).

Regarding step 1(c), Cai discloses assaying the proteins in those subsets using LC-MS/MS with SILAC encoding to characterize and quantify

the relative abundance of the proteins that bind to each type of particle.¹¹ Ex. 1003 ¶¶ 225–229; Ex. 1005, 586–587 (describing “systemic characterization of carbon nanoparticle-protein corona”), 588 (Fig. 3). Cai’s results show “the relative abundance of the MWCNT-binding proteins spread over 8 orders of magnitude” and that “some of the proteins with the highest binding efficiencies” are actually “low abundance proteins” in the lysate, meaning that that Cai’s method selectively enriches such proteins for detection. Ex. 1005, 587. We credit Dr. Vargas’s testimony that these results demonstrate that Cai’s method achieves the “dynamic range” recited in step 1(c). Ex. 1003 ¶¶ 225–229.

Patent Owner’s arguments to the contrary are unavailing. *See* Resp. 21–34. We address each of these arguments in turn.

Patent Owner’s first argument is that the HeLa cell lysate used in Cai is not a “biological sample,” as recited in claim 1, because those cells “are not obtained from an individual subject but are instead vendor-purchased immortalized cells from a cell line cultured in a lab.” Resp. 23.

Patent Owner’s “biological sample” argument is unavailing for several reasons. First, Patent Owner’s argument is premised on its construction of “biological sample,” which we do not adopt for the reasons explained above. *See supra* § III.C.i.

¹¹ To the extent Patent Owner contends that Petitioner’s showing is deficient because it relies on “piecemeal mapping of different aspects of the claim to different experiments in Cai,” we disagree. *See* Resp. 25. As we explained at institution, all of the experiments in Cai are part of the same proteomics study and an ordinarily skilled artisan would understand that the procedures used to form coronas and separate out a subset of proteins in Cai’s Figure 1 are also used to prepare the pulldowns analyzed in Cai’s SILAC analysis. *See* Dec. 18–20 (analyzing disclosure).

Second, even if we adopted Patent Owner’s construction, HeLa cells are a biological sample obtained from a subject—Henrietta Lacks. Ex. 1037, 71 (“The line was derived from cervical cancer cells taken on February 8, 1951 from Henrietta Lacks.”); Ex. 1038, 2 (same); *see also* Ex. 2026, 167 (“HeLa cancer-cell line derived from Henrietta Lacks”). Patent Owner asserts that the HeLa cell line is “a laboratory engineered, immortalized cell line” that is “genetically distinct from whatever was derived from the original donor many decades ago.” Sur-reply 2. Even so, it is undisputed that this cell line originates from cells obtained from a subject (Mrs. Lacks). Ex. 1036, 57:4–11 (Dr. Cima acknowledging “all HeLa cells derive” from Henrietta Lacks). Patent Owner identifies no persuasive reason to exclude them from claim 1, particularly given that the broad description of “[s]uitable biological samples” in the Specification includes both “cell lysates” and “tissue samples” that have been “processed” after being taken from a subject. *See* Ex. 1001, 20:34–39. Accordingly, Cai’s method is performed on a “biological sample” even under the construction Patent Owner proposes for that term.

Patent Owner next contends that Cai does not disclose separating a subset of proteins as recited in step 1(b). Resp. 28–32. According to Patent Owner, claim 1 “require[s] the separated particles [in step 1(b)] to have both (1) different particle types and (2) coronas corresponding to the different particle types comprising overlapping and distinct proteins.” *Id.* at 28–29. Patent Owner urges that Cai does not “disclose[] separating a subset of particles comprising coronas from more than one particle type” because, as shown in Figure 1A, “samples are incubated individually in parallel.” *Id.* at

29–30 (citing Ex. 1003 ¶ 202; Ex. 1005, Fig. 1A; Ex. 2022, 104:1–17, 106:22–107:1, 118:17–20, 119:19–120:9; Ex. 2023 ¶¶ 57–59).

Patent Owner’s step 1(b) argument is unavailing for several reasons. First, this argument is premised on interpreting the phrase “a subset of the plurality of particles comprising the coronas” in step 1(b) in a manner that is contrary to its ordinary meaning. As explained above, that claim language does not require separating particles of each of the multiple types in step 1(a) and can be met by separating any subset of those particles, including a subset that comprises only one particle type. *See supra* § III.C.ii. According to Patent Owner, “the different particle types were always *individually* incubated in Cai’s pull-downs, and that every separation step created a subset of particles comprising a corona from only one particle type.” Resp. 30 (citing Ex. 2022, 104:1–17, 106:22–107:1, 118:17–20, 119:19–120:9 (cross-examination of Dr. Vargas)). Even accepting that characterization, and putting aside the fact that Cai discloses these separation steps were performed in parallel on both particle types, the separation of just one type of particle, e.g., either MWCNT or CB individually, to produce a subset of proteins from that particle type’s coronas, still reads on step 1(b).

Second, Cai discloses the separation of two different particle types comprising coronas with different sets of proteins to produce a subset of proteins from both particle types for subsequent assay. *See* Ex. 1005, 586–587, Fig. 3A (describing preparation of CNT and CB pulldowns). The proteins from each of those pulldowns are then mixed, providing a subset of proteins for the subsequent assay. *Id.* The fact that Cai’s pulldowns are prepared separately by contacting each type of particle with HeLa cell

lysate¹² to form coronas and then eluting proteins bound in those coronas in distinct pulldowns does not distinguish the method of claim 1 because nothing in the claim requires that those steps be carried out in the same reaction vessel or at exactly the same time.¹³

Patent Owner's remaining arguments are all directed to the assaying recited in step 1(c). As explained above, Cai describes assaying the subset of proteins separated in its MWCNT and CB pulldowns using a gel-LC-MS/MS assay that detects the relative abundance (emPAI) of the MWCNT-binding proteins over a range of 8 orders of magnitude. Ex. 1005, 587. Patent Owner argues this does not demonstrate the abundance of those

¹² In its Sur-reply, Patent Owner asserts that Cai's "Fig. 3 experiment used three populations of HeLa cells, and never contacted a single biological sample with the asserted multiple particle types." Sur-reply 7 (quotations omitted). This argument misreads Cai. The same HeLa cell line is lysed to provide the proteins for each pulldown in that experiment. The difference is that the "light," "medium," and "heavy" populations are "metabolically encoded" with "different SILAC media" so that the proteins bound by MWCNT particles can be distinguished from those bound by CB particles in the subsequent mass spectrometry analysis. Ex. 1005, 586. Contrary to Patent Owner's suggestion, an ordinarily-skilled artisan would not understand this encoding to transform those lysates into distinct biological samples. If it did, it would undermine the stated purpose of Cai's experiment, i.e., "identifying the proteins that *selectively* bind to MWCNTs" using "[a] complete protein lysate of HeLa cells . . . as an archetypal human proteome." Ex. 1005, 586 (emphasis added), 591.

¹³ As we noted in the institution decision, the fact that claim 13 expressly recites the performance of a step (i.e., assaying to detect proteins from the coronas of the different particle types) "at the same time" further supports that the steps of claim 1 carry no such requirement. *See* Dec. 25 (quoting Ex. 1001, 87:40–45); *see also Interactive Gift Express, Inc. v. Compuserve Inc.*, 256 F.3d 1323, 1342 (Fed. Cir. 2001) ("Unless the steps of a method [claim] actually recite an order, the steps are not ordinarily construed to require one.").

proteins in the biological sample, but rather their abundance in the pulled-down coronas, i.e., the alleged subset of proteins from step 1(b). Resp. 32–33; Sur-reply 4 (asserting “both parties’ experts agree (and Cai states explicitly), emPAI reflects an estimate of abundance *in the pulled-down corona*” and thus “Cai’s emPAI range is, at best a range “*in the subset*,” not the range in the biological sample”). According to Patent Owner, “the claimed assaying step (element 1[c]) . . . requires detecting ‘proteins *in the biological sample* at concentrations across a dynamic range comprising at least 6 orders of magnitude.’” Sur-reply 1 (emphasis in original).

Petitioner acknowledges that Cai’s stated 8 orders of magnitude range refers to measurements on the subset of proteins in Cai’s pulldowns, but asserts that is what step 1(c) requires. That is, Petitioner contends “the claim language expressly requires ‘*assaying the subset* of proteins of (b) with an instrument to detect, *in the subset*’ proteins in the sample at the required concentration range. Thus[,] the actions claimed in limitation 1(c) occur with respect to the subset, not the biological sample.” Reply 19; *see also* Tr. 8:7–9:8 (explaining Petitioner’s position the dynamic range in step 1(c) refers to the concentrations of proteins in the subset of step 1(b)).¹⁴

¹⁴ To the extent Patent Owner contends that Petitioner waived this argument, we disagree. *See* Ex. 2031, 9 (Patent Owner demonstrative). Petitioner presented exactly this interpretation of the claim language in the Petition. *See* Pet. 60 (“A POSA would have understood that the relative abundance of bound proteins ‘spread over 8 orders of magnitude’ shows that Cai’s MS analysis detected protein concentrations *in the subset* across a dynamic range comprising the claimed ‘*at least 6 orders of magnitude*’”) (first emphasis added)).

These competing arguments therefore turn on a question of claim interpretation—does the “dynamic range” in step 1(c) refer to the concentration range in the subset of proteins of step 1(b), or instead, as Patent Owner contends, to the range in the original biological sample?

We agree with Petitioner that the recited “dynamic range” refers to concentrations in the subset of step 1(b). Step 1(c) recites: “*assaying the subset of proteins of (b) with an instrument to detect, in the subset, proteins in the biological sample at concentrations across a dynamic range comprising at least 6 orders of magnitude.*” Ex. 1001, 86:66–87:3 (emphases added). Because the claim language specifically states that both the assaying and detecting are performed on the subset of proteins from step 1(b), the most natural reading of step 1(c) is that “concentrations across a dynamic range at least 6 orders of magnitude” also refers to that subset.

In addition to the claim language, Petitioner’s position is supported by the Specification, which asserts that Patent Owner’s “sensor array” has a larger dynamic range than that of “current assays” such as “mass spectrometry.” Ex. 1001, 20:20–29. That is, the Specification refers to the “dynamic range” of the instruments used to do the assaying, which step 1(c) makes clear is conducted on the subset of proteins from step 1(b).

Extrinsic evidence similarly indicates that the term “dynamic range” is understood in the art to refer to the “range over which a technique has a measurable response to the presence of *an analyte*.” Ex. 1003 ¶ 44; 1032, 3 (emphasis added). The analyte in step 1(c) is the subset of proteins from step 1(b). *See* Ex. 1001, 86, 66 (“[1](c) assaying the subset of proteins of [1](b) with an instrument”).

In contrast, Patent Owner’s interpretation is unavailing because it relies on parsing the language of step 1(c) to shift focus away from the “subset” and redirect it to the phrase “in the biological sample.” *See* Sur-reply 1 (omitting the entire first half of step 1(c) from its quotation of the claim language). Such an approach is improper because it does not consider that phrase in context of the surrounding claim language. *See Phillips v. AWH Corp.*, 415 F.3d 1303, 1314 (Fed. Cir. 2005) (en banc) (explaining that the “surrounding words of the claim . . . must be considered”) (internal quotations omitted). In context, the better reading is that “in the biological sample” refers to the source of the proteins in the subset, whereas the “dynamic range” refers to the sensitivity of the assay to detect proteins from the sample that are present in the subset at concentrations across a range of at least 6 orders of magnitude. For these reasons, Patent Owner’s argument that Cai only demonstrates an 8-order of magnitude detection range for the subset of step 1(b) (i.e., the proteins separated in the MWCNT and CB pulldowns) is not persuasive.

Patent Owner also asserts that Petitioner’s showing is deficient because “emPAI is not a measure of concentration.” Resp. 32. Patent Owner points to Cai’s description of emPAI as “a *semi-quantitative* indication of the *overall abundance* of protein *in the mixture*.” *Id.* (quoting Ex. 1005, 587) (Patent Owner’s emphasis). According to Patent Owner, “abundance is not the same as concentration.” *Id.* (citing Ex. 2023 ¶¶ 63–64). Patent Owner further suggests that the emPAI values Petitioner relies upon in Cai are inaccurate because they “are not found within the narrow range that had been validated for this procedure.” *See id.* at 33–34 (citing Ex. 2023 ¶ 66).

Patent Owner's emPAI arguments are unavailing for several reasons. First, we credit Dr. Vargas's testimony that a POSA would understand that "a dynamic range of 6 orders of magnitude is a relative concentration range between high abundant proteins and low or rare abundant proteins" and that "the emPAI values of detected proteins disclosed by Cai that span from 0.02 to 2254683 are relative concentration ranges that are at least 6 orders of magnitude. Ex. 1003 ¶ 229 (citing Ex. 1005, 12–17 (Supplementary Table 2)).¹⁵ Dr. Vargas's testimony is supported by Ishihama, which Cai cites to explain the emPAI values in its LC-MS/MS analysis. Ex. 1005, 587. Ishihama shows that emPAI is directly correlated with and can be mathematically converted to protein concentration. Ex. 1018, 1268 (explaining that emPAI "is directly proportional to protein content" and can be used to "calculate the absolute concentrations"), Fig. 1(E) (showing linearity between emPAI and "Protein concentration (fmol/μL)" over tested range). Dr. Vargas shows that when Cai's largest and smallest emPAI values are converted using Ishihama's formula, the resulting molar percent values continue to span 8 orders of magnitude. Ex. 1003 ¶¶ 239–240. Thus, the record supports that emPAI and concentration are sufficiently related such that a POSA would understand that Cai's disclosure of emPAI values spanning 8 orders of magnitude is a dynamic concentration range of "at least 6 orders of magnitude" as recited in step 1(c).

¹⁵ Table S2 is part of Cai's "[s]upplementary data," which can be obtained at a URL referenced in Cai. Ex. 1012 ¶ 33 (quoting Ex. 1005, 592). Patent Owner does not dispute that this supplementary data is part of Cai's disclosure for purposes of Petitioner's patentability challenges.

Second, it is no moment that Cai only reports emPAI values because no particular measure of protein concentration is required by claim 1. Indeed, claim 1 does not require any quantification of the concentration of proteins detected in step 1(c). *See* Tr. 36:14–37:1 (Patent Owner’s counsel acknowledging that claim 1 does not require concentration quantification to occur). All that is required is that the subset be assayed to “detect” proteins present at concentrations spanning at least 6 orders of magnitude—a threshold that Cai’s span of emPAI values exceeds by a very wide margin. Thus, even accepting that there are differences between abundance and concentration, the differences identified by Dr. Cimas (*see* Ex. 2023 ¶ 64) are not significant enough to undermine Dr. Vargas’s testimony above.

Third, Patent Owner and Dr. Cimas’s assertions about the relationship between emPAI and concentration not being validated for over the full range of values in Cai are overstated. Both Ishihama and Cai indicate that emPAI is a reliable measure both for detecting and quantifying the detected proteins in a proteomic assay. Ex. 1018, 1268 (“the emPAI approach seems to provide a reasonably accurate estimate for comprehensive absolute quantification”), 1271 (suggesting “routine use” of emPAI “in reporting of proteomic results”); *see also* Ex. 1005, 587 (adopting Ishihama’s emPAI approach for its study). Indeed, Dr. Cimas’s testimony confirms that Isikawa “validated” the linear relationship between emPAI and protein concentration with empirical data spanning a significant range. *See* Ex. 2023 ¶ 66 (concentration tested over range from 30 fmol/μL to 1.8 pmol/μL). Dr. Cimas observes that Cai’s emPAI values span a wider range than those tested in Ishihama, but does not identify any reason why one of ordinary skill in the art would not expect the correlation shown in Ishihama to

continue through the full extent of Cai’s wider range. *Id.* Thus, Patent Owner’s arguments based on that testimony are just speculation that we do not find persuasive to overcome Petitioner’s showing for step 1(c).

In sum and based on the full trial record, Petitioner has shown by a preponderance of the evidence that Cai anticipates claim 1.

ii. Claims 4, 6, 17, 22, and 25

Each of these claims depends directly from claim 1. Claims 4, 6, and 17 recite limitations relating to the assaying in step 1(c). Ex. 1001, 87:11–16 (assaying performed “using gel-electrophoresis, liquid chromatography, mass spectrometry” and other techniques), 20–21 (“dynamic range of the proteins detected spans 6 or more orders of magnitude”), 54–55 (“assaying further comprises digesting the proteins”). Claim 22 limits the recited “biological sample” to a Markush group of options that includes “any lysates.” *Id.* at 88:12–17. Claim 25 further recites that the “different particle types” in step 1(a) differ “by one or more physicochemical properties” including “composition, size, surface charge, hydrophobicity, hydrophilicity, surface functionality, surface topography, surface curvature, shape, and any combination thereof.” *Id.* at 88:31–37.

Petitioner provides argument, supported by evidence, explaining that Cai also discloses the additional limitations in claims 4, 6, 17, 22, and 25. *See* Pet. 60–61. Patent Owner does not dispute Petitioner’s showing for any of these claims separately from the arguments already addressed above for claim 1. *See* Resp. 34 (asserting that “the petition fails to establish Cai anticipates claim 1” and thus the challenge to these dependent claims “similarly fails”).

We adopt Petitioner's undisputed analysis of the additional limitations of claims 4, 6, 17, 22, and 25, and credit its supporting evidence. *See* Pet. 60–61; Ex. 1003 ¶¶ 94–97, 199–200, 206, 230–235. Accordingly, Petitioner has shown by a preponderance of the evidence that Cai anticipates these claims.

F. Obviousness over Cai

Petitioner contends claims 1, 4, 6, 17, 22, and 25 are obvious over Cai. *See* Pet. 62–64. This ground relies on the same showing as in the anticipation ground with some additional obviousness contentions relating to step 1(c). *Id.* As before, Patent Owner disputes Petitioner's contentions. *See* Resp. 35–36.

We find that Petitioner has shown by a preponderance of the evidence that these claims would also have been obvious over Cai. To the extent this ground presents the same issues and arguments as the anticipation ground, we have addressed those above. *See supra* § III.E. Below, we focus on Petitioner's additional contentions for step 1(c) and Patent Owner's arguments against the same.

i. Claim 1

Petitioner contends that “[t]o the extent that Cai is found not to disclose limitation 1[c] regarding the orders of magnitude . . . Cai would still render obvious claims 1, 4, 6, 17, 22, and 25.” Pet. 62. Petitioner relies on calculations by Dr. Vargas, who uses equations disclosed in Ishihama, to convert the largest and smallest emPAI values in Cai's Table S2 into “more traditional concentration numbers,” i.e., molar percent or “mol %” values. *Id.* at 63 (citing Ex. 1003 ¶¶ 239–240; Ex. 1018, 1267 (conversion formula)).

Petitioner further asserts that “a POSA would have been motivated to search out protein concentration,” and could use the equations disclosed in Ishihama to confirm the concentration of proteins detected using Cai’s method. *Id.*; *see also* Tr. 47:13–48:13 (explaining argument).

We find Petitioner’s showing above credible. That is, Cai discloses emPAI values for proteins detected in its LC-MS/MS analysis that span 8 orders of magnitude. Ex. 1005; *see also* Ex. 1003 ¶ 229 (“emPAI values of detected proteins disclosed by Cai . . . span from 0.02 to 2254683”). Relying on Ishihama’s Equation 3, Dr. Vargas converts the highest and lowest emPAI values into molar percent values and observes that those “more traditional concentration numbers” likewise span 8 orders of magnitude. Ex. 1003 ¶¶ 239–240.

Neither Patent Owner, nor Dr. Cimas, dispute the accuracy of Dr. Vargas’s calculations. *See* Resp. 35–36. Instead, Patent Owner relies on Dr. Cimas’s testimony that “[m]olar percent (mol %) of protein content is indicative of relative abundance; it is not a measure of concentration.” Ex. 2023 ¶ 69. But, other than Ishihama itself, Dr. Cimas does not cite any support for his assertion that a POSA would not understand molar percent to be a measure of concentration. *See id.* This is a problem for Patent Owner because Ishihama expressly refers to the values calculated using its formulae as “absolute concentrations.”¹⁶ Ex. 1018, 1268. In other words, there is

¹⁶ Ishihama refers to “absolution concentrations . . . calculated using Equation 4.” Ex. 1018, 1268. Equation 4 calculates “Protein content (weight %)” from emPAI. Ex. 1018, 1267. The same characterization of protein content as a measure of concentration would apply to Equation 3, which likewise calculates “Protein content” from emPAI, but as a molar percent. *Id.*

competing expert testimony on this point—but Dr. Vargas’s testimony that molar percent is a measure of concentration (Ex. 1003 ¶ 239) is supported by the other evidence of record (Ishihama), whereas Dr. Cimas’s is not. For this reason, we credit this aspect of Dr. Vargas’s testimony over the competing testimony from Dr. Cimas.

Dr. Cimas’s suggestion that Ishihama’s conversion formulae are inaccurate is also unavailing. *See* Ex. 2023 ¶ 69 (referring to “observed error as high as three-fold the detected values”). First, contrary to Dr. Cimas’s suggestion, Ishihama specifically states that its “emPAI-based concentrations,” i.e., concentrations calculated from measured emPAI values using its formulae, are “highly consistent with the actual values.” Ex. 1018, 1268. Second, as explained above, claim 1 only requires that proteins be detected over the recited dynamic range; it does not require the quantification of those proteins. Moreover, this dynamic range is a relative range and Cai teaches that its method detects proteins at a concentration over a much larger dynamic range (8 orders of magnitude) than the “at least 6 orders” recited in claim 1. Ex. 1003 ¶ 229. Accordingly, even if some error would have been expected in the mathematical conversion of emPAI to molar percent, the record does not suggest that this error would be significant enough to undermine Petitioner’s showing that the dynamic range of detection achieved by Cai’s method meets the “at least six orders of magnitude” threshold required by claim 1.

Indeed, this would seem to be true both for the protein concentration in Cai’s pull-downs (i.e., the subset of step 1(b)) as well as the concentration in the underlying biological sample. Cai teaches that the binding of proteins to its MWCNT and CB particles is “protein-selective” and that “[s]ome of

the proteins with the highest binding efficiencies . . . are low abundance proteins” in the HeLA lysate. Ex. 1005, 587. Thus, Cai teaches that these particles are able to “selectively enrich” proteins that may be of low abundance in the sample so that they may be detected by assaying the protein subset from the MWCNT and CB-pull downs. *Id.* at 592. The broad eight orders of magnitude range of protein concentration values detected in Cai’s subset suggests that its assay is sensitive enough to detect both high and low abundant proteins and would be capable of detecting proteins present in a similar dynamic range of concentrations in the underlying biological sample. *See* Ex. 1003 ¶ 228 (“A POSA would have understood that these results from assaying the biological sample using MS across a dynamic range large enough to detect both low abundant and high abundant proteins in the biological sample, as disclosed in Cai.”), 241 (“[I]t was well-known and desirable in the field (and in fact, a necessity when assessing datasets from research such as the proteins detected in Cai’s method) to look up the proteins concentration to determine the orders of magnitude of the dynamic range for the detected low abundant and high abundant proteins.”).

For these reasons, and those set forth above (*supra* § III.E.i.), Petitioner has shown by a preponderance of the evidence that claim 1 would have been obvious over Cai.

ii. Claims 4, 6, 17, 22, and 25

There are no separate obviousness contentions offered for the additional limitations recited in dependent claims 4, 6, 17, 22, and 25. Pet. 62–63. Petitioner apparently relies on the same showing in its anticipation ground for those limitations. *See id.* As before, Patent Owner

does not dispute Petitioner’s showing for the dependent claims separately from the arguments addressed above for claim 1.

Accordingly, for the same reasons discussed above, Petitioner has shown by a preponderance of the evidence that claims 4, 6, 17, 22, and 25 would have been obvious over Cai. *See supra* § III.E.ii.

G. Obviousness over Hu

Petitioner contends claims 1–4, 6, 7, 17, and 22–25 would have been obvious over Hu. Pet. 21–46. Patent Owner disputes those contentions. *See* Resp. 36–53.

For the reasons explained below, we find that Petitioner has not shown that claims 1–4, 6, 7, 17, and 22–25 would have been obvious over Hu. More specifically, Petitioner has not sufficiently shown that Hu teaches or suggests step (c) of claim 1. Because this issue is dispositive for all of the claims challenged in this ground, our analysis below focuses on the parties’ arguments and evidence pertaining to step 1(c).

Petitioner argues that Hu teaches contacting a biological sample (blood plasma) with different particle types (Fe₃O₄ NPs of varying size) to form protein coronas, separating a subset of proteins from those coronas, and then using “‘RPLC’ (*i.e.*, Reverse Phase Liquid Chromatography) in conjunction with tandem mass spectrometry (RPLC-MS/MS)” to assay the proteins in those coronas. *See* Pet. 21–33. According to Petitioner, Hu discloses using software to “search the known International Protein Index [IPI] for MS/MS spectral data protein matches.” *Id.* at 33 (citing Ex. 1003 ¶ 155; Ex. 1008, 356).

Petitioner acknowledges that Hu does not disclose the dynamic range of its method in numeric form, but asserts that it would have been obvious

that Hu’s method achieves a dynamic range of at least six orders of magnitude based on its detection of two specific proteins—serum albumin and LTBP-1. Pet. 35–36. Hu identifies serum albumin as one of the “most abundant plasma proteins bound on the Fe₃O₄ NPs.” *Id.* at 34 (quoting Ex. 1008, Table 2). LTBP-1 is one of more than a hundred other proteins listed in Hu’s supplementary data table. *Id.* (citing Ex. 1008, 18 (IPI entry “IPI00973177.1”). Petitioner relies on what it calls “reference values” in other exhibits, to assert that “serum albumin is normally present in healthy human plasma at concentrations ranging from 35 to 52 mg/mL and LTBP-1 in concentrations ranging from 1.7 to 6.0 ng/mL.” *Id.* at 36 (citing Ex. 1015, 786–87, Table 2 (serum albumin); Ex. 1016, 16046–48, Table 1 (LTBP-1). Based on those reference values, Petitioner concludes that “the method of Hu detects the bound proteins in the biological sample at a concentration greater than 10 mg/mL (10^{-2}) (*i.e.*, serum albumin) and less than 10 ng/mL (10^{-9}) (*i.e.*, LTBP-1), comprising a dynamic range of at least six orders of magnitude.” *Id.* at 36 (citing Ex. 1003 ¶¶ 166, 103–104).

Petitioner’s showing is deficient in multiple respects. First, Petitioner’s showing is premised on reference values that purportedly show the concentrations of serum albumin and LTBP-1 normally present in human blood plasma (*i.e.*, the biological sample), as opposed to their concentration in the subset of proteins separated from the coronas on the Fe₃O₄ NPs (*i.e.*, the subset of step 1(b)). As explained above, for the Cai-based grounds Petitioner takes the position that the dynamic range in step 1(c) refers to the concentrations in the subset of step 1(b) and we agree with that interpretation. *See supra* § III.E.i. Even if we were to credit Petitioner’s evidence that the concentration of serum albumin and LTBP-1 in human

blood plasma typically differs by at least six orders of magnitude, it would not show that the concentrations of those proteins in the subset separated from the coronas on Hu's Fe_3O_4 NPs also differ by at least six orders of magnitude. Thus, Petitioner has not shown that a POSA would have understood Hu's method to be sensitive enough to detect proteins in the subset at concentrations over a dynamic range of at least six orders of magnitude, as recited in step 1(c).

Second, we agree with Patent Owner that Petitioner's evidence does not show that there is an at least six orders of magnitude difference in the typical concentration of the proteins Petitioner relies upon in its showing for step 1(c). *See* Resp. 47–49. Hu indicates that it was “LTBP1 isoform 4” that was detected in its spectral data.¹⁷ Ex. 1008, 18 (IPI entry “IPI00973177.1”). But the evidence Petitioner offers as a “reference value” for that protein's concentration is not for isoform 4, but rather a different isoform, isoform 1. *Compare* Ex. 1016, 16051 (“There are four isoforms of LTBP1s (LTBP-1, -2, -3, -4)” *with* 16048 (Table 1 showing “LTBP-1” concentration)). Accordingly, Petitioner's evidence does not show the concentration of protein corresponding to the spectral data it relies upon.

Moreover, Patent Owner offers evidence from another source suggesting that the typical concentration of LTBP1 isoform 4 in human blood plasma is much higher than the value Petitioner relies upon. Ex. 2027, 1 (reporting an estimated concentration of 220 ng/ml). As Dr. Cimas

¹⁷ Patent Owner asserts that “a POSA would be highly skeptical that Hu actually detected LTBP-1” because the signal data Hu reports for that protein is “extremely small” and not observed for all replicates. Resp. 44–45. We need not resolve that issue because even assuming the data Petitioner relies upon is not experimental error, Petitioner's showing still fails.

explains, using this higher reference value there are only 5 orders of magnitude difference between the concentration of serum albumin and the particular LTBP1 protein detected in Hu. Ex. 2023 ¶¶ 98–99. We credit this testimony because it is supported by the record. Indeed, Petitioner offers no particular response, either to the failure of its own evidence to show a concentration for LTPB1 isoform 4 or to Patent Owner’s evidence suggesting a higher concentration for that protein, in the Reply. Accordingly, even assuming the recited “dynamic range” could be shown based on typical concentrations of proteins in the biological sample, Petitioner has not shown that those concentrations extend over a broad enough range for the proteins detected by Hu’s method.

Third, to the extent Petitioner attempts to rely on general assertions regarding the sensitivity of “known mass spectrometry” techniques to supplement the teachings in Hu, those arguments are unpersuasive. *See* Pet. 37–38 (arguing the ’360 Patent “admits that known mass spectrometry can detect proteins at a dynamic range of 4-6 order[s] of magnitude” and “there is nothing inventive . . . by claiming a dynamic range that was already well known and understood”); Reply 24 (urging that “Patent Owner should not be heard to argue that” mass spectrometry and shotgun methods “fail to provide sufficient sensitivity and specificity for use with the patented method *because the patent says they do*”). We do not doubt that mass spectrometry techniques sensitive enough to achieve a dynamic range of at least six orders of magnitude were known in the art.¹⁸ But Petitioner does

¹⁸ Indeed, as explained above, Cai discloses a prior art method that uses mass spectrometry to detect proteins at concentrations over a dynamic range that exceeds six orders of magnitude. *See supra* § III.E–F.

not assert that it would have been obvious to modify Hu to use one of these other known techniques. *See generally* Pet. 33–39. Rather, Petitioner’s ground is premised on the use of Hu’s particular RPLC-MS/MS assay and Petitioner has not demonstrated that Hu’s assay teaches or suggests the recited dynamic range.

For these reasons, Petitioner has not shown by a preponderance of the evidence that claim 1 would have been obvious over Hu. Petitioner’s showing for dependent claims 2–4, 6, 7, 17, and 22–25 relies upon the same showing for the limitations incorporated from base claim 1. Accordingly, Petitioner has not met its burden to show that these claims would have been obvious over Hu for the same reasons as claim 1.

H. Obviousness over Zaccaria

Petitioner contends claims 1, 2, 4, 6, 7, 17, 22, 23, and 25 would have been obvious over Zaccaria. Pet. 64–80. Patent Owner disputes those contentions. *See* Resp. 53–65.

For the reasons explained below, we find that Petitioner has not shown that claims 1, 2, 4, 6, 7, 17, 22, 23, and 25 would have been obvious over Zaccaria. More specifically, Petitioner has not sufficiently shown that Zaccaria teaches or suggests step (c) of claim 1. Because this issue is dispositive for all of the claims challenged in this ground, our analysis below focuses on the parties’ arguments and evidence pertaining to step 1(c).

Petitioner argues that Zaccaria teaches contacting a biological sample (RBC lysate) with different particle types (superparamagnetic NPs) to form protein coronas around the NPs, separating a subset of proteins from those coronas, and then using “nano-LC-MS/MS protein identification” to assay the proteins in those coronas. *See* Pet. 64–73. Similar to its contentions for

the ground based on Hu, Petitioner acknowledges that “Zaccaria does not specifically state the dynamic range in numerical form,” but instead points to particular proteins identified in Zaccaria’s assay as evidence of its dynamic range. *Id.* at 74. That is, Petitioner relies on Zaccaria’s identification of “Hemoglobin subunit beta and Hemoglobin subunit alpha (HBA1)” as amongst the “twenty most abundant proteins identified from NPs corona” and “myotubularian-related protein 5 (SBF1)” as an example of a lower abundant protein allegedly identified through Zaccaria’s method. *Id.* at 73–74 (citing Ex. 1007, 1875–76 (Table 2 identifying HPA1), 58 (supplementary data table)). Petitioner relies on another reference, as evidence that an ordinarily-skilled artisan “would have understood that HBA1 is normally present in RBC lysate in a concentration of 1.47×10^4 pmol/mg and SBF1 in a concentration of 6.62×10^{-6} pmol/mg because both proteins are well-known components of RBC.” *Id.* at 74–75 (citing Ex. 1009, 33, 34). In addition, for claim 7, Petitioner relies on a similar analysis for another lower abundant protein, “PIK3R4.” *See id.* at 79. Petitioner alleges that the range between the normally present concentrations of these proteins in RBC lysate teaches or suggests the “dynamic range” recited in step 1(c). *Id.* at 75.

Petitioner’s showing is deficient in multiple respects. First, it suffers from the same defect as the ground based on Hu, i.e., Petitioner does not show that Zaccaria’s assay detects proteins present in the subset separated from the NP coronas at concentrations that differ by at least six orders of magnitude. Rather, Petitioner attempts to show that Zaccaria’s method can detect proteins present in the biological sample (RBC lysate) at concentrations that vary by the recited six orders of magnitude. But the

concentration range in the RBC lysate is not the same as the range in the subset separated from the NP coronas. As Petitioner observes, Zaccaria teaches that “low-abundant proteins identified in nontreated sample [i.e., the RBC lysate] were present in higher abundance on NPs” and at the same there was a “respective decrease of high-abundant proteins” on those NPs. Pet. 75 (quoting Ex. 1007, 1881). This results in a “greatly reduce[d] range of concentrations in the [NP treated] sample” that was assayed by mass spectrometry. *Id.* Petitioner has not shown that this reduced range comprises at least six orders of magnitude. Accordingly, even if we credit Petitioner’s evidence as showing a greater than six orders of magnitude difference between the typical concentrations of these proteins in RBC lysate, Petitioner’s evidence does not show that Zaccaria teaches or suggests that its assay is sensitive enough to detect proteins present in the subset of step 1(b) at concentrations across the recited dynamic range.

Second, we agree with Patent Owner that Petitioner has not demonstrated that SBF1 and PIK3R4 were actually detected in Zaccaria’s assaying step. As Dr. Cimas explains, an ordinarily-skilled artisan would understand that Zaccaria’s mass spectrometry analysis involves degrading the proteins “into small peptide fragments, acquir[ing] spectral data, and analyz[ing] the raw mass spectral data to make ‘predictions’ of the ‘theoretical peptides possible.’” Ex. 2023 ¶ 114 (quoting Ex. 1003 ¶ 39). Zaccaria does this by mapping the data to “different accession numbers (e.g., different numbers identifying difference proteins in a database).” *Id.* ¶ 112; *see also* Ex. 1007, 24–118 (Zaccaria’s supplemental data tables listing accession numbers (AC) for spectral data). Zaccaria indicates that, in some instances, the spectral data for particular peptide fragments will match

to several different proteins and “if several proteins match the same set of peptides, only one member of the protein group is indicated in [its] concise lists.” Ex. 1007, 17; Ex. 2023 ¶ 117. Petitioner relies on lists identifying the accession numbers for SBF1 and PIK3R4 for certain spectral data. *See* Pet. 73–74 (citing Ex. 1007, 58 (listing AC identifier O95248 for SBF1); Ex. 1003 ¶¶ 119, 270–271, 281, 284. However, as Dr. Cimas credibly demonstrates (Ex. 2023 ¶¶ 114–118), Zaccaria elsewhere indicates that the same data also matches to other proteins having different accession numbers and that Petitioner offers no evidence for the concentration range of these other proteins. The fact that neither Petitioner in its Reply, nor Dr. Vargas in her declaration, confront this issue undermines the credibility of Petitioner’s showing for step 1(c). *See* Reply 25–26; Ex. 1003 ¶¶ 119, 270, 281 (failing to address the multiple accession number issue for the spectral data they contend shows Zaccaria’s detection of SBF1 and PIK3R4).

Finally, Petitioner’s argument that “there is nothing inventive in claim 1 by claiming a dynamic range that was admittedly already well known and understood” is unpersuasive. *See* Pet. 76 (referring back to Petitioner’s argument that the Specification acknowledges that mass spectrometry was already known to achieve a dynamic range of up to six orders of magnitude). As was the case with the Hu-based ground, Petitioner does not assert that it would have been obvious to modify Zaccaria to use a different or more sensitive assay. Instead, the Petition relies on Zaccaria’s existing nano-LC-MS/MS assay, but Petitioner has not sufficiently demonstrated that assay teaches or suggests the recited dynamic range.

For these reasons, Petitioner has not shown by a preponderance of the evidence that claim 1 would have been obvious over Zaccaria. Petitioner’s

showing for dependent claims 2, 4, 6, 7, 17, 22, 23, and 25 relies upon the same showing for the limitations incorporated from base claim 1.

Accordingly, Petitioner has not met its burden to show that any of the challenged claims would have been obvious over Zaccaria.

IV. CONCLUSION¹⁹

Petitioner has shown that claims 1, 4, 6, 17, 22, and 25 are anticipated by and would have been obvious over Cai. Petitioner has not shown that claims 1–4, 6, 7, 17, and 22–25 would have been obvious over Hu.

Petitioner has also not shown that claims 1, 2, 4, 6, 7, 17, 22, 23, and 25 would have been obvious over Zaccaria.

Claims	35 U.S.C. §	Reference(s)/ Basis	Claims Shown Unpatentable	Claims Not shown Unpatentable
1–4, 6, 7, 17, 22–25	103	Hu		1–4, 6, 7, 17, 22–25
1, 4, 6, 17, 22, 25	102	Cai	1, 4, 6, 17, 22, 25	
1, 4, 6, 17, 22, 25	103	Cai	1, 4, 6, 17, 22, 25	
1, 2, 4, 6, 7, 17, 22, 23, 25	103	Zaccaria		1, 2, 4, 6, 7, 17, 22, 23, 25

¹⁹ Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner’s attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. See 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. See 37 C.F.R. § 42.8(a)(3), (b)(2).

Claims	35 U.S.C. §	Reference(s)/ Basis	Claims Shown Unpatentable	Claims Not shown Unpatentable
Overall Outcome			1, 4, 6, 17, 22, 25	2, 3, 7, 23, 24

V. ORDER

Accordingly, it is:

ORDERED that Petitioner has shown that claims 1, 4, 6, 17, 22, and 25 of U.S. Patent 11,435,360 B2 are unpatentable;

ORDERED that Petitioner has not shown that claims 2, 3, 7, 23, and 24 of U.S. Patent 11,435,360 B2 are unpatentable;

FURTHER ORDERED that, because this is a Final Written Decision, parties to this proceeding seeking judicial review of our Decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

IPR2024-01473
Patent 11,435,360 B2

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